Practice workbook · When is the particle at rest (or at a given velocity)?

Setting velocity (or acceleration or displacement) equal to a value and solving for t.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-26

1. Question 5 · part a · 3 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part a only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

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1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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2. Question 3 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

3 A particle P moves along the x-axis.

At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 6t - 16$

When $t = 0$, P is at the origin and is moving with velocity 12 m/s.

(a) Find an expression in terms of t for

(i) the velocity of P at time t seconds

(ii) the displacement of P at time t seconds.

(4)

(b) Hence find the time at which P first returns to the origin.

(3)

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2. Question 3 · part b · 3 marks

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Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 7 marks)

2. Question 3 · part b · 3 marks · Working and answer

Working and answer

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3. Question 4 · part b · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

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3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

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1 A particle P is moving in a straight line. At time t seconds, $t \geqslant 0$, the displacement, s metres, of P from a fixed point O of the line is given by

$$s = 3 + 8t + t^2 - \frac{1}{3}t^3$$

Find the distance of P from O when P is instantaneously at rest.

(4)

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(Total for Question 1 is 4 marks)

4. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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5. Question 5 · part b · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

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5. Question 5 · part b · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

4 A particle P is moving along a straight line. At time t seconds $(t \geqslant 0)$, its velocity, v m/s, is given by $v = 4t^2 - 6t + 5$

(a) Find the minimum speed of P

(3)

The acceleration of P at time T seconds is $18\ \text{m/s}^2$

(b) Find the value of T

(2)

When $t = 0$, P is at the point X, and when $t = 3$, P is at the point Y

(c) Find the distance XY

(4)

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6. Question 4 · part b · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 5 · part a · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

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8. Question 5 · part a · 3 marks · Working and answer

Working and answer

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9. Question 1 · part whole · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part whole only.

1 A particle P is moving along a straight line that passes through the fixed point O.
At time t seconds, $t \geqslant 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 4t^2 - 27t + 4$$

Find the value of t at the instant when the velocity of P is 8 m/s.

(4)

(Total for Question 1 is 4 marks)

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9. Question 1 · part whole · 4 marks · Working and answer

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10. Question 6 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

(c) Find the value of t for which P is accelerating at $10\,\text{m/s}^2$ in the positive x direction.

(3)

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10. Question 6 · part b · 4 marks · Working and answer

Working and answer

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11. Question 6 · part c · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

(c) Find the value of t for which P is accelerating at $10\,\text{m/s}^2$ in the positive x direction.

(3)

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11. Question 6 · part c · 3 marks

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Question 6 continued

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(Total for Question 6 is 8 marks)

11. Question 6 · part c · 3 marks · Working and answer

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12. Question 2 · part b · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

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12. Question 2 · part b · 2 marks · Working and answer

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13. Question 3 · part a · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

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13. Question 3 · part a · 3 marks · Working and answer

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14. Question 3 · part b · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

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14. Question 3 · part b · 3 marks · Working and answer

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15. Question 5 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

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15. Question 5 · part b · 2 marks · Working and answer

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16. Question 5 · part b · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

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16. Question 5 · part b · 2 marks · Working and answer

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